# COMPUTER NETWORKS - UE24CS252B
# Lab 1: Designing and Simulation of Network Topology using Cisco Packet Tracer

**Objectives:**

· To understand the purpose of Cisco Packet Tracer.

· To navigate, choose network and end devices and customize them.

· To interconnect devices and configure them using a simple interface.

· To become familiar with building topologies in Packet Tracer.

· To simulate data interactions traveling through a network.

**Prerequisites:**

This lab assumes some understanding of the building blocks of communication networks and the internet. At this point, we haven't discussed other protocols but you may use Packet Tracer in later labs to discuss those as well. Several types of devices and network connections can be used. For this experiment we will keep it simple by using end devices, switches, routers, and connections.

Steps to install Cisco Packet Tracer:

1. **Go to the download page**:
   Visit this link: Download Cisco Packet Tracer
2. **Download Packet Tracer**:
   Scroll down and download the version compatible with your operating system (Windows or Mac or Linux).
3. **Install Packet Tracer**:
   Once the download is complete, run the installer and follow the on-screen instructions to complete the installation.
4. **Disable your internet connection**:
   Before launching Packet Tracer for the first time, turn off your internet connection. This will prevent it from asking for registration.
5. **Launch Packet Tracer**:
   Open the installed Cisco Packet Tracer. Since you are offline, it will allow you to use the program without requiring registration.

**Steps to install Cisco Packet Tracer on Mac OS:**

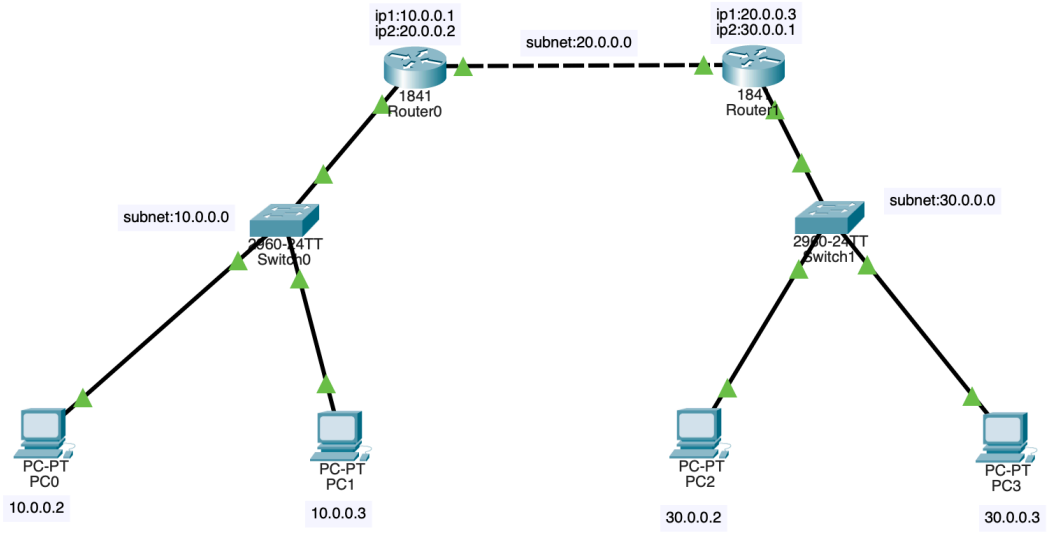
6. **Go to the download page**:
   Visit this link: Download Cisco Packet Tracer
7. **Download Packet Tracer**:
   Scroll down and download the version compatible with your OS.
8. **Install Packet Tracer**:
   Once the download is complete, open the .dmg file. Drag the Cisco Packet Tracer application into the Applications folder as instructed.
9. **Adjust mac OS Security Settings (if required):**
   On first launch, mac OS may block the app.Go to **System Settings → Privacy & Security**.Under Security, click Open Anyway for Cisco Packet Tracer.
10. **Disable your internet connection**:
    Before launching Packet Tracer for the first time, turn off your internet connection. This will prevent it from asking for registration.
11. **Launch Packet Tracer**:
    Open **Applications → Cisco Packet Tracer.** Since you are offline, it will allow you to use the program without requiring registration.

# Task 1 - Demo

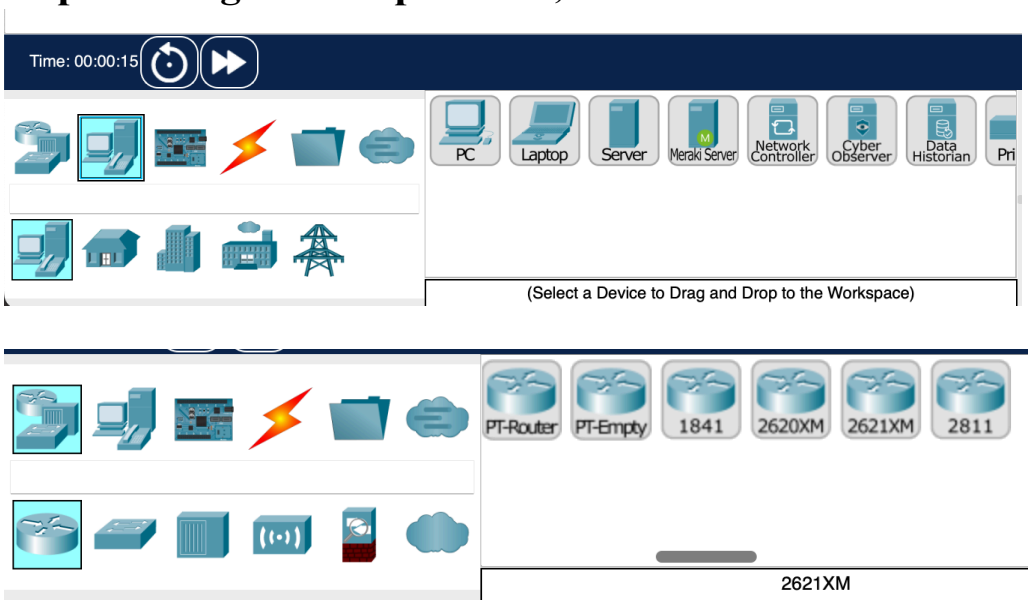
## Demo 1

### Network Topology:

To replicate the given scenario, create a topology in packet tracer, as shown in the following image.

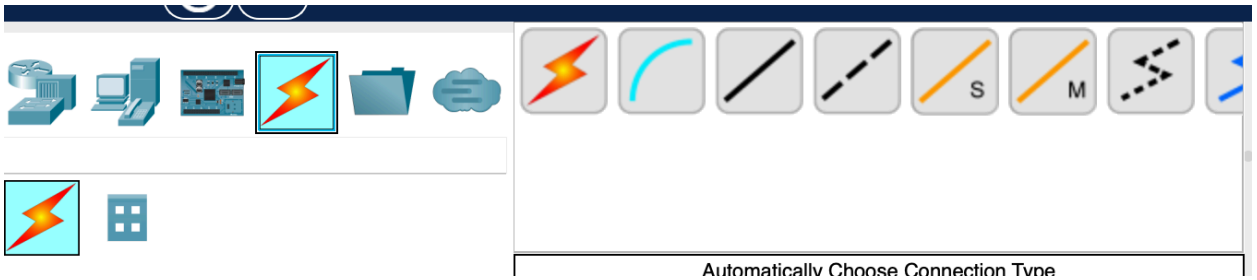



### Step 1 - Drag and drop the PC, router and switch.




### NOTE: Use only 1841 router for labs unless mentioned

**Step 2 - Now connect them using wire (the first one) according to the topology above.**




## Step 3 - Network setup

### PC & Router Configuration Details:

**PC0:** IP Address ---> 10.0.0.1
Gateway ---> 10.0.0.3

**PC1:** IP Address ---> 10.0.0.2
Gateway ---> 10.0.0.3

**PC2:** IP Address ---> 30.0.0.2
Gateway ---> 30.0.0.1

**PC3:** IP Address ---> 30.0.0.3
Gateway ---> 30.0.0.1

Check if it is the right port for which you are adding the IP address for the router by toggling the **Port status button.**

**Router 0:**

FastEthernet0/0 ---> 10.0.0.3

FastEthernet0/1 ---> 20.0.0.1

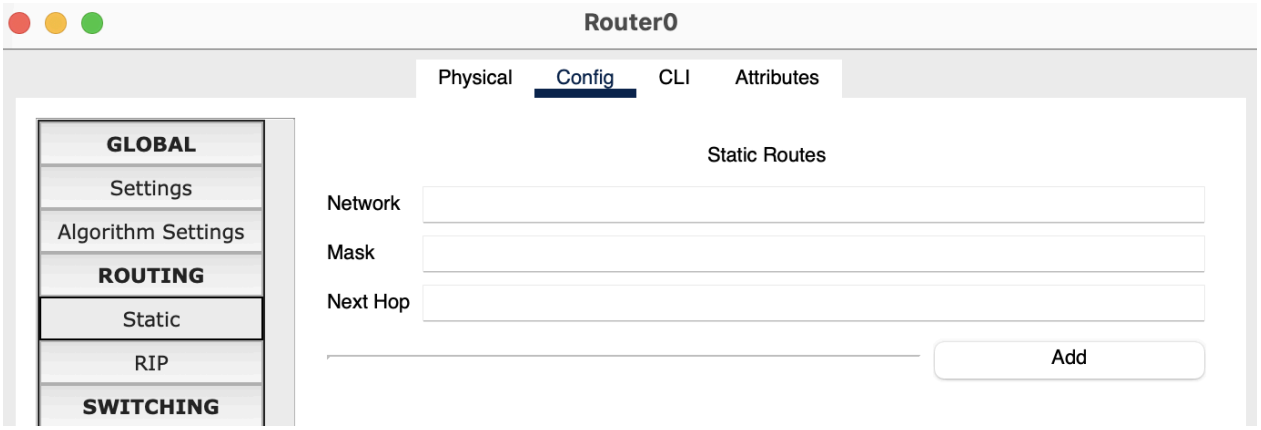
**Router 1:**

FastEthernet0/0 ---> 20.0.0.2

FastEthernet0/1 ---> 30.0.0.1

## Step 4 - Routing Table Entries:

**Router Network Next Hop (Go to Routing->Static)**




Router0
**Network:** 30.0.0.0
**Mask:** 255.0.0.0
**Next Hop:** 20.0.0.2

Router1
**Network:** 10.0.0.0
**Mask:** 255.0.0.0
**Next Hop:** 20.0.0.1

**Execution Procedure:**

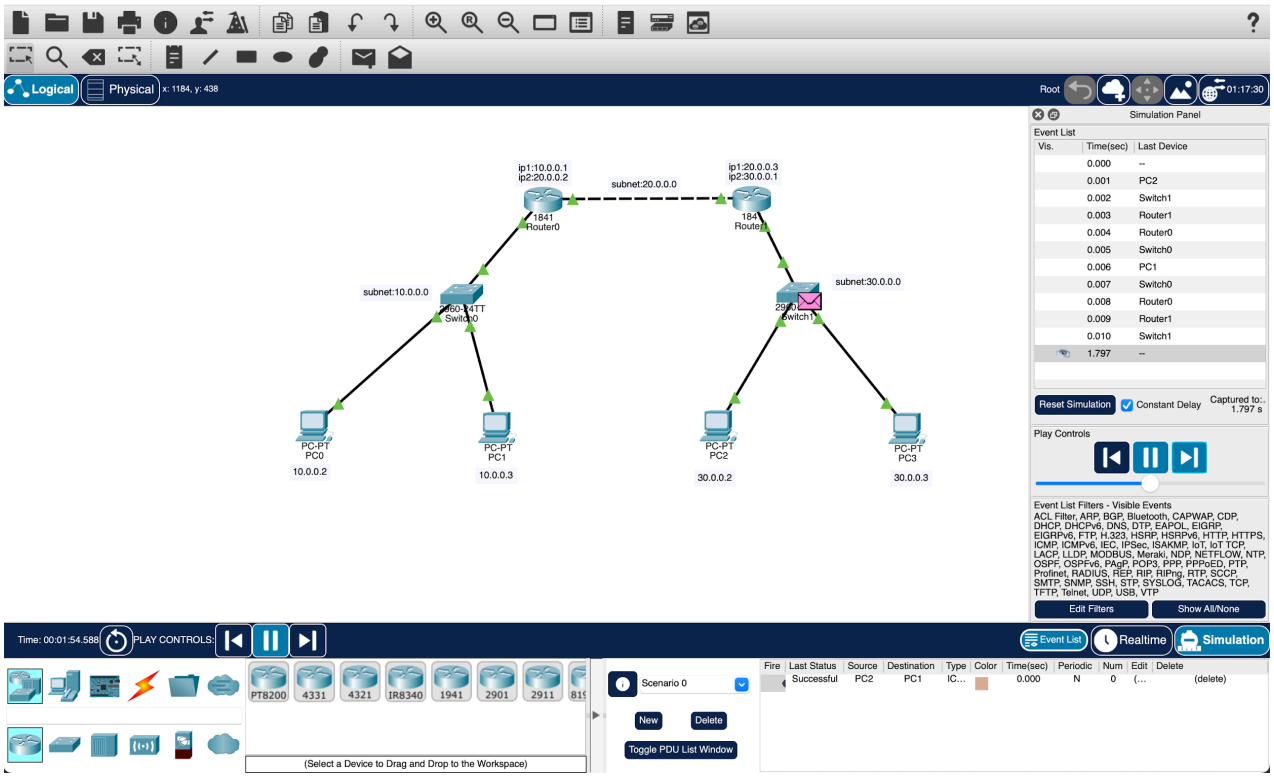
**Step 1:** Design a network topology with desktops, switches and routers similar to the network depicted in the above diagram.

**Step 2:** Configure the PCs and routers with the details provided above.

**Step 3:** Send a simple PDU from any PC on network 10.0.0.0 to any other PC on other network 30.0.0.0 and vice-versa.

**Step 4:** Simulate the network and observe the packet flow from one network to another.

**Figure of the topology, simulation hop table and the table where successful transmission is seen**



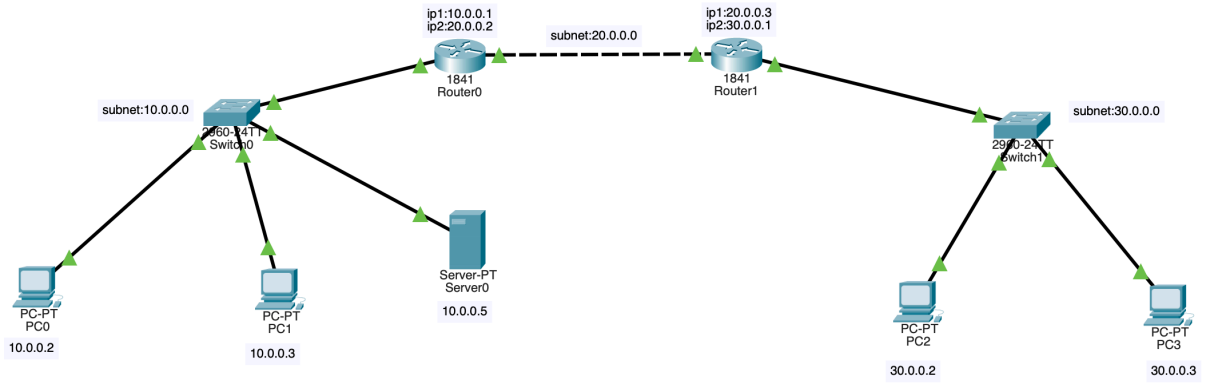

# Demo 2

**Implementation of a Local DNS Server using Cisco Packet Tracer Objectives:**

- To understand the purpose of DNS Server.
- To implement the Local DNS server by selecting appropriate devices and customizing them.
- To simulate data interactions traveling through a network.
- To demonstrate ping operation with IP as well as the name of the host.

## Network Topology

To replicate the given scenario, create a topology in packet tracer, as shown in the following image. Using the topology from Demo 1, add DNS server at Switch0 and configure as below.




## Step 1 - DNS Configuration Details:

Here along with the IP we'll assign the DNS server address to let the PC know about its Local DNS Server

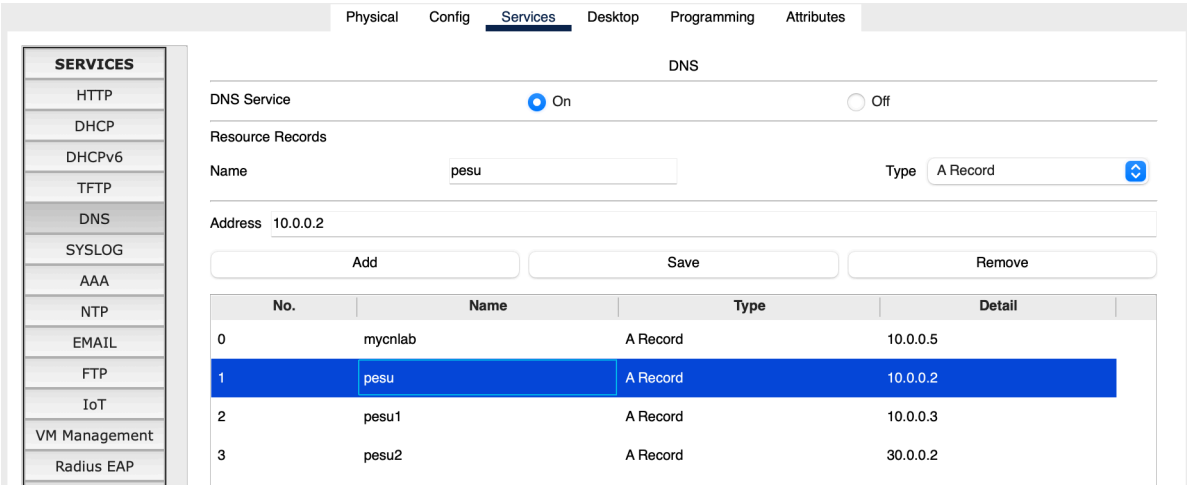
| | |
|---|---|
| **Server** | **IP - 10.0.0.5**<br>**DNS - 10.0.0.5** |

**Step 1a** - Add 10.0.0.5 in **DNS server config in global settings** on all the PCs.

## Step 2 - Adding DNS entries

### Step 2a - Click on the server- >services->DNS –> DNS service =on

Name = PESU
IP= 10.0.0.2
**ADD**



Click on PC2 - 30.0.0.2
**Desktop->Command Prompt ->** ping 10.0.0.2

```
C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time=1ms TTL=126
Reply from 10.0.0.2: bytes=32 time=1ms TTL=126
Reply from 10.0.0.2: bytes=32 time=1ms TTL=126
Reply from 10.0.0.2: bytes=32 time<1ms TTL=126

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

**Step 2b - Click on the server- >services->DNS –> DNS service =on**

Name = PESU
IP= 10.0.0.2
**ADD**

Click on other PC2 - 30.0.0.2
**Desktop->Command Prompt ->** ping pesu

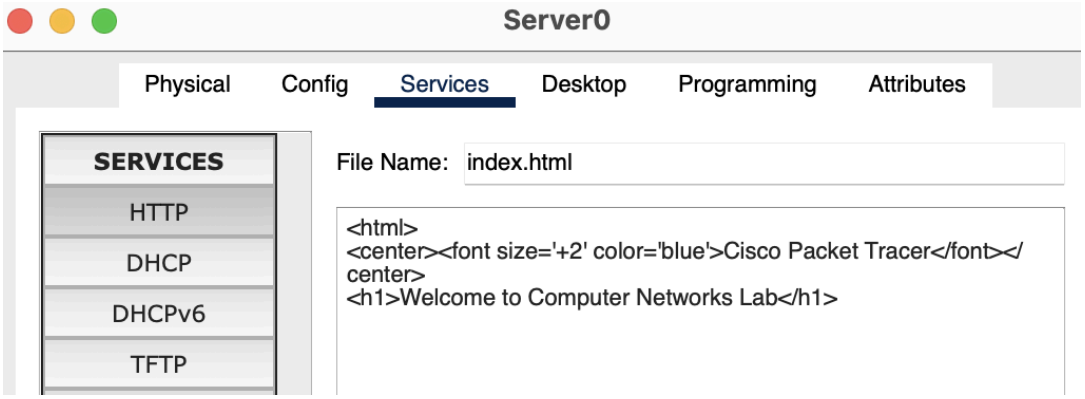
```
C:\>ping pesu

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time<1ms TTL=126
Reply from 10.0.0.2: bytes=32 time<1ms TTL=126
Reply from 10.0.0.2: bytes=32 time<1ms TTL=126
Reply from 10.0.0.2: bytes=32 time<1ms TTL=126

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Click on **Web Server->Services-> HTTP --> Index.html**edit.




Save. (Overwrite= yes)

**Click on the server- >services->DNS –> DNS service =on**

Name : Mycnlab
IP : 10.0.0.5
Click ADD

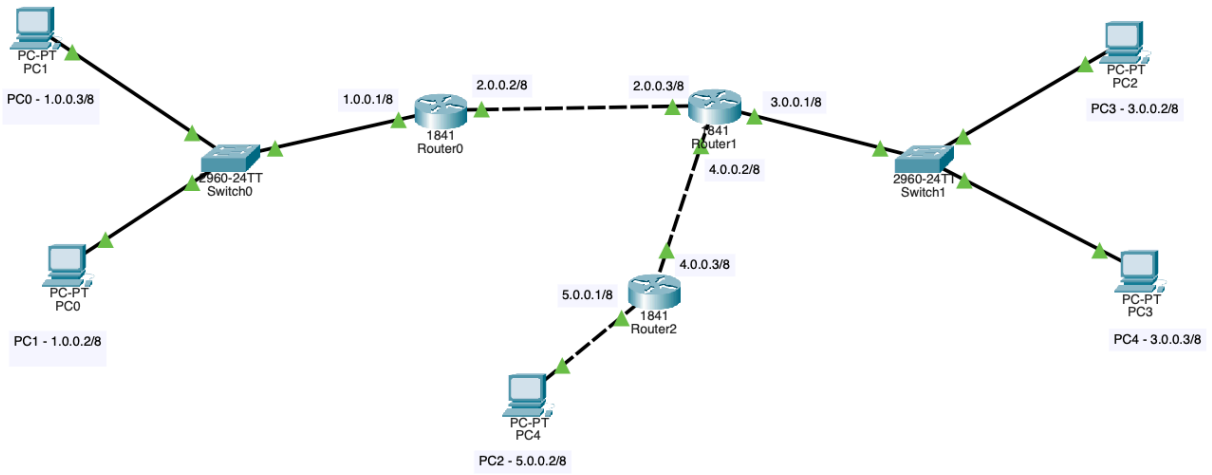
Click on **PC2->Display->web browser** type Mycnlab in browser and you should be able to access the HTML contents. Also any PC should be able to ping any other PC in the network by IP and Name Both.



# Task 2 (Mandatory)

## Task 2a

In this task, students should explore how to add interfaces to the router. Connect all routers, switches and hosts with their IP addresses as shown below.

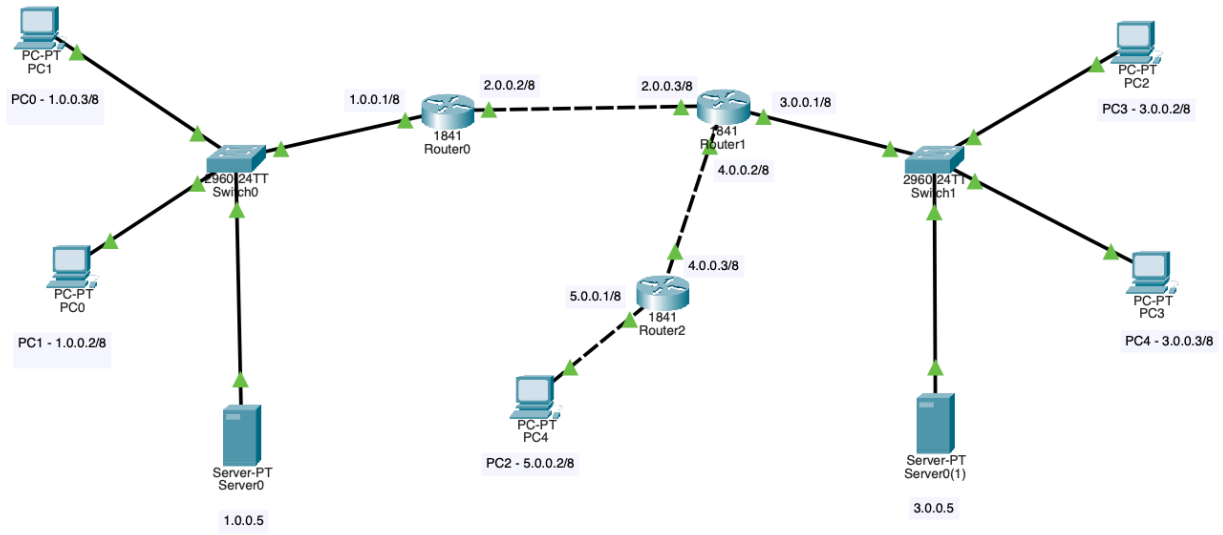



## Deliverables/Submission:

1. Screenshot of the working topology including the **successful transmission of packet** with your **SRN in place holder.**
2. Simulation hop table
3. What is your observation and understanding of this concept that you have implemented?
4. Why is it necessary to assign an IP address to the newly added interface?
5. What is a routing table and why is it important?

# Task 2b

Students should create the given topology and get the successful ping by adding entries in the DNS Server. Students should also be able to access the web server.




## Deliverables/Submission:

1. Screenshots of the working implementation of the above topology with both the **DNS servers in working condition** with your **SRN in placeholder.**
2. What is your observation and understanding of this concept that you have implemented?
3. Why must each PC be configured with the DNS server's IP address?
4. What is the role of a **DNS server** in a network?

## Conclusion (Final Submission)

- A PDF document with all the **mandatory tasks (Task 2a & 2b).**
- **Naming convention:** SRN_NAME_LAB1
- Observations and answers to all the questions for all these tasks are mandatory.
- All the screenshots must be clear, if not marks will not be allotted.
- Every component should have a placeholder name with its IP address.
- Your **SRN in placeholder** is mandatory.
- Any plagiarism found will be **straight away rejected** and the **lab will be awarded 0 marks**. **Be careful.**